

MRUNMAYEE LIMAYE

[github](#) | [linkedin](#) | mrunmayee.limaye.01@gmail.com

EDUCATION

Visvesvaraya National Institute of Technology (VNIT), Nagpur
Bachelor of Technology (B.Tech.) in Computer Science and Engineering

Aug 2023 – Present
CGPA: 8.70/10

EXPERIENCE

Natwest

May 2026 - Jul 2026

Summer SDE Intern | [\[Certificate\]](#)

- Investigated excessive production logging in AWS CloudWatch, traced root causes to the Java/Spring Boot codebase, and implemented logging optimizations merged into release branches, eliminating 10K unnecessary logs per day.
- Collaborated in an Agile development team using GitLab and JIRA to develop backend features, contribute to code reviews, and maintain production-ready code through version control workflows.

IIT Roorkee

May 2025 - Jul 2025

Summer Research Intern | [\[Certificate\]](#)

- Applied a reinforcement learning policy for autonomous quadcopter control with a suspended payload.
- Built a 200 Hz data collection and processing pipeline for real-time policy deployment on quadcopter hardware, generating a 3K+ sample dataset across 50+ flight experiments.

IvLabs

May 2024 - Oct 2024

Summer Intern | [\[GitHub\]](#)

- Built a 4-wheeled gesture-controlled omnidirectional robot for real-time multi-axis navigation.
- Programmed motion control using ESP32, MPU6050, ESP-NOW, and PWM to support 10 wireless navigation commands. Presented the project at the 3rd International Conference on PCEMS 2024.

PUBLICATIONS

- “Prompted to Fly: Translating Free-Form Instructions into Schema-Constrained Mission Generation for UAVs using LLMs” *IEEE BigData 2025*, pp. 7278–7286. [doi: 10.1109/BigData66926.2025.11401432](#).

PROJECTS

CareFlow Clinic Management System | [\[GitHub\]](#)

Apr 2026

- Engineered a full-stack clinic management system with role-based access control (RBAC) supporting 3 user roles.
- Designed a normalized relational database with 10 tables and PL/SQL triggers to automate billing workflows and maintain scheduling integrity.
- Created analytics dashboards for patient records, appointment management, and operational reporting.

SAFE: Structured Agentic Feedback Engine | [\[GitHub\]](#)

Mar 2026

- Developed a multi-agent Automated Program Repair (APR) pipeline using AST-based code analysis and LLMs for automated localized code repair.
- Engineered syntax-preserving patch integration, achieving 100% syntactically valid code generation.
- Benchmarked against naïve prompting on QuixBugs, improving repair success while reducing patch size by 37%.

Autonomous Multi-UAV Search & Rescue System | [\[GitHub\]](#)

Nov 2025 - Jan 2026

- Created an end-to-end real-time UAV autonomy pipeline for human detection, GPS localization, mission planning, and autonomous payload delivery using computer vision and flight-controller telemetry.
- Developed distributed multi-UAV coordination using inter-drone communication, dynamic path planning, and autonomous mission execution validated through 20+ autonomous flight tests.
- Integrated computer vision with autonomous navigation to create a real-time perception-to-action pipeline achieving less than 1m localization error.

SKILLS

Languages: Java, Python, C++, C, SQL

Frameworks & Libraries: Spring Boot, FastAPI, React.js, PyTorch, OpenCV, REST APIs, MAVLink, DroneKit, ArduPilot

Databases: PostgreSQL, Oracle SQL, MongoDB, MySQL

Cloud & DevOps: Linux, JIRA, AWS CloudWatch, Docker, Git, GitLab